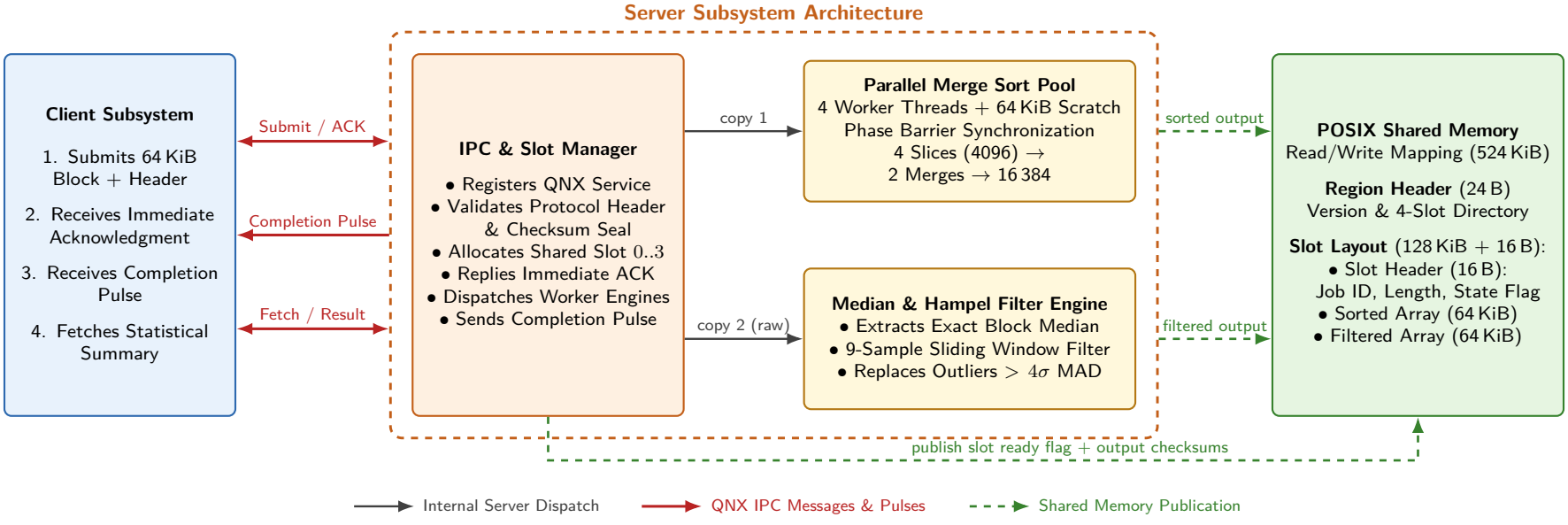

Server Subsystem Architecture
Client Subsystem
1. Submits 64 KiB Block + Header
2. Receives Immediate Acknowledgment
3. Receives Completion Pulse
4. Fetches Statistical Summary
Submit / ACK
Completion Pulse
Fetch / Result
IPC & Slot Manager
• Registers QNX Service
• Validates Protocol Header & Checksum Seal
• Allocates Shared Slot 0..3
• Replies Immediate ACK
• Dispatches Worker Engines
• Sends Completion Pulse
copy 1
copy 2 (raw)
Parallel Merge Sort Pool
4 Worker Threads + 64 KiB Scratch
Phase Barrier Synchronization
4 Slices (4096) →
2 Merges → 16 384
Median & Hampel Filter Engine
• Extracts Exact Block Median
• 9-Sample Sliding Window Filter
• Replaces Outliers > 4σ MAD
sorted output
filtered output
POSIX Shared Memory
Read/Write Mapping (524 KiB)
Region Header (24 B)
Version & 4-Slot Directory
Slot Layout (128 KiB + 16 B):
• Slot Header (16 B):
Job ID, Length, State Flag
• Sorted Array (64 KiB)
• Filtered Array (64 KiB)
publish slot ready flag + output checksums
Internal Server Dispatch
QNX IPC Messages & Pulses
Shared Memory Publication